

# Technical Annex

## Interpreting Outputs from the DEA, SFA, Time-Trend, and Malmquist Efficiency Tool

*User guide for analysts, managers, and reviewers*

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### **Purpose of this annex**

This annex explains the major outputs produced by the efficiency tool. It is intended to accompany exported results and to help users interpret DEA efficiency, SFA efficiency, DEA targets and peers, scale diagnostics, time trends, Malmquist productivity indices, bootstrap intervals, and solver diagnostics.

### **Important scope note**

The tool estimates cross-sectional DEA and pooled half-normal SFA models. The Time Trends tab summarizes changes in estimated scores over time. The Malmquist tab estimates DEA-based adjacent-period productivity change. These outputs are related, but they answer different questions.

### How this annex is organized

- 1. Core concepts and model setup
- 2. Data preparation and analysis-unit construction
- 3. DEA outputs: efficiency, targets, peers, slacks, and scale diagnostics
- 4. SFA outputs: frontier coefficients, efficiency, variance, and fit diagnostics
- 5. Comparing DEA and SFA results
- 6. Time Trends outputs
- 7. Malmquist productivity outputs
- 8. Bootstrap, robustness, and sensitivity outputs
- 9. Diagnostics, limitations, and recommended reporting language
- 10. Output column dictionary and glossary

## 1. Core concepts and model setup

The tool evaluates the relative performance of decision making units, abbreviated as DMUs. A DMU can be a hospital, clinic, county, provider group, school, facility, program, or any comparable unit that uses inputs to produce outputs.

The core idea is frontier analysis. A frontier is an estimated best-practice boundary. Units on the boundary receive best-practice scores; units inside the boundary have estimated room for improvement relative to the observed or estimated frontier.

| Concept          | Meaning in this tool                                                                                                   | Examples                                                                                                    |
|------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| DMU              | The unit being evaluated. If both a DMU column and time column are selected, the app builds DMU-period analysis units. | County in 2014, county in 2022, hospital-year, clinic-quarter                                               |
| Inputs           | Resources, costs, or capacity measures to reduce or use more efficiently.                                              | FTEs, beds, costs, expenditures, staffing, capital                                                          |
| Outputs          | Services, outcomes, or production measures to maintain or increase.                                                    | Visits, discharges, treated cases, quality-adjusted output                                                  |
| Orientation      | The direction of improvement in DEA.                                                                                   | Input-oriented: reduce inputs. Output-oriented: expand outputs.                                             |
| Returns to scale | The assumption about how output changes when all inputs change proportionally.                                         | CRS: proportional scaling. VRS: scale-adjusted frontier.                                                    |
| Efficiency score | A normalized summary of distance from the frontier. For standard DEA and bounded SFA, 1 is best.                       | DEA_Efficiency = 0.84 means an estimated 16% radial improvement opportunity under the selected orientation. |

### 1.1 What each method answers

| Method      | Question answered                                                                                                                                        | Main outputs                                                                            |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| DEA         | How far is each DMU from an observed best-practice production frontier under the selected inputs, outputs, orientation, and returns-to-scale assumption? | DEA_Efficiency, targets, peers, slacks, scale efficiency, RTS, benchmark support        |
| SFA         | How far is each DMU from a parametric stochastic frontier after separating statistical noise from one-sided inefficiency?                                | SFA_Efficiency, coefficients, sigma_u, sigma_v, gamma, AIC/BIC, convergence diagnostics |
| Time Trends | Did each unit's already-estimated efficiency score improve or decline across observed time periods?                                                      | Period summaries, first-to-last changes, improvement shares, trajectories               |
| Malmquist   | Did productivity change between adjacent periods, and was the change due to catch-up or frontier shift?                                                  | Malmquist_MPI, Efficiency_Change_EC, Technical_Change_TC, chain indices                 |

**Interpretation rule of thumb**

For DEA\_Efficiency, Scale\_Efficiency, most SFA\_Efficiency settings, and Malmquist\_MPI, values closer to 1 or above 1 where applicable are better. Always check whether the selected SFA cost convention reports a bounded score or a cost ratio.

## 2. Data preparation and analysis-unit construction

The app preserves selected metadata, coerces model variables to numeric values, and removes rows with missing, non-numeric, or non-positive model values. Positive values are required because DEA envelopment models and log-linear SFA models require non-negative or strictly positive data; SFA requires strictly positive values unless the variables are explicitly marked as already logged.

### 2.1 DMU and time columns

When both a DMU identifier and time column are selected, the app treats each row as a DMU-period observation. For example, Facility A in 2014 and Facility A in 2022 are separate analysis units, not duplicate records. Duplicate handling is only applied when the same unit appears more than once within the same selected time period.

```
Example analysis-unit labels
Facility A | 2014
Facility A | 2022
Facility B | 2014
Facility B | 2022
```

### 2.2 Group or peer-set filtering

A group column can be used to restrict the analysis to comparable units. This is important because DEA is relative to the selected peer set. Mixing fundamentally different service types, geographies, or case-mix groups may create misleading frontiers.

### 2.3 Raw and logged variables

DEA should generally use raw positive levels. SFA may use raw positive variables, in which case the app logs them internally, or already logged variables, in which case the app uses the variables as selected. This distinction is important when matching R or Stata models that explicitly use  $\ln\_y$  and  $\ln\_x$  variables.

## 3. DEA outputs

Data Envelopment Analysis is a nonparametric linear-programming approach. It constructs an empirical frontier from the observed input-output combinations and compares each DMU with a convex combination of peer DMUs.

### 3.1 DEA model solved by the tool

For a target DMU 0 with inputs  $x_0$  and outputs  $y_0$ , the tool solves an envelopment model. Under variable returns to scale (VRS), the constraint  $\sum(\lambda_{bdj}) = 1$  is added. Under constant returns to scale (CRS), that convexity constraint is omitted.

| Orientation        | Optimization problem in words                                                                                                                    | Reported score                                                                                                                                    |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Input-oriented DEA | Find the largest proportional input contraction that still allows the unit to produce at least its observed outputs using the selected reference | DEA_Efficiency = $\theta$ , where $0 < \theta \leq 1$ . A score of 0.80 means a 20% radial input reduction opportunity before considering slacks. |

| Orientation         | Optimization problem in words                                                                                  | Reported score                                                                                                                                                                         |
|---------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | technology.                                                                                                    |                                                                                                                                                                                        |
| Output-oriented DEA | Find the largest proportional output expansion that is feasible using no more than the unit's observed inputs. | The LP factor is $\phi \geq 1$ . The app reports $\text{DEA\_Efficiency} = 1 / \phi$ so 1 remains best. The output expansion factor is also exported as $\text{DEA\_Radial\_Factor}$ . |

Input-oriented DEA:  
 minimize  $\theta$   
 subject to  $X \lambda \leq \theta * x_0$   
 $Y \lambda \geq y_0$   
 $\lambda \geq 0$   
 $\sum(\lambda) = 1$  for VRS

Output-oriented DEA:  
 maximize  $\phi$   
 subject to  $X \lambda \leq x_0$   
 $Y \lambda \geq \phi * y_0$   
 $\lambda \geq 0$   
 $\sum(\lambda) = 1$  for VRS

### 3.2 Main DEA score columns

| Column                         | Definition                                                                                                      | How to interpret                                                                                                                                                               |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DEA_Efficiency                 | Primary 0-1 DEA score. For input orientation it equals $\theta$ . For output orientation it equals $1 / \phi$ . | 1 means the DMU is on the selected DEA frontier. Lower values indicate greater relative inefficiency.                                                                          |
| DEA_Efficiency_0_1             | Alias of DEA_Efficiency retained for clarity and exports.                                                       | Use this when you want a bounded DEA score regardless of orientation.                                                                                                          |
| DEA_Radial_Factor              | The raw radial factor from the DEA LP: $\theta$ for input orientation or $\phi$ for output orientation.         | For input orientation, lower than 1 implies input reduction opportunity. For output orientation, greater than 1 implies output expansion opportunity.                          |
| DEA_R_Benchmarking_Style_Score | Compatibility score for packages that report output orientation as $\phi$ rather than $1 / \phi$ .              | Useful when validating against R Benchmarking or similar outputs. For output orientation, smaller rank values can correspond to better performance when $\phi$ is closer to 1. |
| DEA_Rank                       | Rank based on DEA_Efficiency, with higher bounded scores ranked better.                                         | Rank 1 is the best bounded DEA score. Ties are possible.                                                                                                                       |
| DEA_Status and DEA_Message     | Linear-programming solver status and message.                                                                   | Check these columns if scores are missing or unexpected.                                                                                                                       |

### 3.3 DEA targets, slacks, and improvement opportunities

DEA targets translate scores into operational quantities. Targets are calculated from the peer-weighted projection  $X \lambda$  and  $Y \lambda$ . Slacks are non-radial adjustments that remain after the proportional radial movement.

| Output group          | Examples of columns                                             | Meaning                                                                                                |
|-----------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Current values        | Current_Input_<name>,<br>Current_Output_<name>                  | Observed input or output value used in the DEA run.                                                    |
| Target values         | Target_Input_<name>,<br>Target_Output_<name>                    | Peer-implied target value on the selected frontier.                                                    |
| Total input reduction | Total_Input_Reduction_<name>,<br>Total_Input_Reduction_%_<name> | Estimated reduction from current input to DEA target input. Especially central in input orientation.   |
| Total output increase | Total_Output_Increase_<name>,<br>Total_Output_Increase_%_<name> | Estimated increase from current output to DEA target output. Especially central in output orientation. |
| Non-radial slacks     | NonRadial_Input_Slack_<name>,<br>NonRadial_Output_Slack_<name>  | Additional input/output adjustment beyond the radial efficiency movement.                              |
| DEA_Peers             | Text list of peer DMUs and lambda weights                       | The reference units whose convex combination defines the target.                                       |

#### How to use target tables

Targets are model-based benchmarks, not automatic operating mandates. Before turning a target into a management goal, check data quality, case-mix comparability, omitted outputs, and whether the peer set is operationally realistic.

### 3.4 Peer weights and benchmark support

Peer weights, also called lambda weights, identify which reference DMUs form the comparison point for each evaluated DMU. A peer with a large lambda weight contributes more to the target.

| Column                  | Definition                                                      | How to interpret                                         |
|-------------------------|-----------------------------------------------------------------|----------------------------------------------------------|
| Peer_DMU_ID             | A reference unit used in the target DMU's DEA projection.       | A DMU can have multiple peers.                           |
| Lambda_Weight           | Weight assigned to the peer in the convex or conic combination. | Larger values indicate stronger influence on the target. |
| Reference_Count         | Number of evaluated DMUs for which this unit appears as a peer. | High values identify commonly used benchmarks.           |
| Reference_Weight_Sum    | Total lambda weight received across evaluated DMUs.             | Captures both frequency and weight intensity.            |
| Benchmark_Support_Score | Reference_Count scaled by the maximum observed reference count. | A relative benchmark prominence score from 0 to 1.       |

### 3.5 CRS, VRS, scale efficiency, and returns to scale

The app estimates CRS and VRS DEA scores and uses their ratio to summarize scale efficiency. CRS asks whether a unit is efficient when scale size is also part of the benchmark. VRS asks whether the unit is efficient after allowing for scale differences.

| Column             | Definition                                                    | How to interpret                                                                                                                                 |
|--------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| DEA_CRS_Efficiency | DEA score under constant returns to scale.                    | Measures technical plus scale efficiency.                                                                                                        |
| DEA_VRS_Efficiency | DEA score under variable returns to scale.                    | Measures pure technical efficiency after relaxing the scale assumption.                                                                          |
| Scale_Efficiency   | DEA_CRS_Efficiency / DEA_VRS_Efficiency where defined.        | Values near 1 indicate little scale-related inefficiency. Values below 1 suggest scale mismatch.                                                 |
| CRS_Lambda_Sum     | Sum of CRS reference-set lambda weights.                      | Used as a diagnostic for returns to scale.                                                                                                       |
| Returns_to_Scale   | Categorical diagnostic: CRS, IRS, DRS, Ambiguous, or Unknown. | IRS suggests operating below most productive scale size; DRS suggests operating above it. Treat as diagnostic when multiple optima are possible. |

### 3.6 Super-efficiency and cross-efficiency

| Output                    | Definition                                                                                     | Use                                                                                                               |
|---------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| DEA_SuperEfficiency       | Leave-one-out DEA score for each DMU. The evaluated DMU is removed from its own reference set. | Ranks frontier units that all have DEA_Efficiency = 1. Can exceed 1 or be infeasible depending on model and data. |
| DEA_SuperEfficiency_Rank  | Rank based on super-efficiency.                                                                | Useful for distinguishing among efficient frontier DMUs.                                                          |
| CCR_Cross_Efficiency_Mean | Mean score assigned to each DMU by all DMUs' multiplier weights under a CRS/CCR model.         | A peer-appraisal robustness measure. It is not the same as benchmark support.                                     |
| Self_vs_Cross_Gap         | DEA_Efficiency minus mean cross-efficiency.                                                    | Large gaps may indicate that a unit looks efficient mainly under its own favorable weights.                       |

## 4. SFA outputs

Stochastic Frontier Analysis estimates a parametric frontier while explicitly allowing for random noise. This is useful when measurement error, unobserved shocks, or statistical noise may explain some distance from the frontier.

### 4.1 SFA model forms

| Specification       | Model form              | Interpretation                                                                               |
|---------------------|-------------------------|----------------------------------------------------------------------------------------------|
| Production frontier | $\ln(y) = f(x) + v - u$ | Output can fall below the frontier because of inefficiency $u \geq 0$ and random noise $v$ . |
| Cost frontier       | $\ln(c) = f(x) + v + u$ | Cost can exceed the frontier because of inefficiency $u \geq 0$ and random noise $v$ .       |

| Specification | Model form                                      | Interpretation                                                                                       |
|---------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Cobb-Douglas  | $f(x) = \beta_0 + \sum \beta_i \ln(x_i)$        | First-order coefficients are elasticities when variables are logged.                                 |
| Translog      | Cobb-Douglas plus squared and interaction terms | Allows elasticities and substitution patterns to vary with input levels. Requires more observations. |

#### Single-output SFA

The SFA module is single-output. If multiple DEA outputs are selected, the app asks which output should be used for SFA. DEA can continue using multiple outputs in the same run.

## 4.2 Main SFA score and ranking columns

| Column                         | Definition                                                                               | How to interpret                                                                                                                                          |
|--------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| SFA_Efficiency                 | Conditional efficiency derived from the estimated half-normal inefficiency distribution. | For production frontiers and the default cost convention, values are bounded between 0 and 1 and 1 is best.                                               |
| SFA_Rank                       | Rank based on SFA_Efficiency.                                                            | For bounded scores, rank 1 is highest. For Stata/Frontier cost-ratio convention, rank 1 is lowest because 1 is best and values above 1 imply excess cost. |
| SFA_Model_Input_Columns        | Input variables used in the SFA model.                                                   | May differ from DEA inputs if separate SFA variables were selected.                                                                                       |
| SFA_Model_Output_Column        | Single dependent variable used in SFA.                                                   | Check this when DEA has multiple outputs.                                                                                                                 |
| SFA_Variables_Already_Logged   | Whether selected SFA variables were treated as logged.                                   | TRUE means the app did not log the variables again.                                                                                                       |
| SFA_Cost_Efficiency_Convention | Efficiency convention used for cost frontiers.                                           | Default bounded reciprocal score or Stata/Frontier-style cost ratio.                                                                                      |

## 4.3 SFA coefficient and model-fit outputs

| Output                        | Meaning                                                            | Interpretation                                                                       |
|-------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Coefficient                   | Estimated beta parameter in the Cobb-Douglas or Translog frontier. | In Cobb-Douglas production models, first-order coefficients are output elasticities. |
| Std_Error                     | Approximate standard error from the optimizer Hessian.             | Use cautiously; bootstrap intervals may be more informative when available.          |
| t_stat and p_value            | Approximate Wald test for coefficient = 0.                         | Screening statistics, not a substitute for a fully specified inferential workflow.   |
| CI_Lower_95% and CI_Upper_95% | Approximate 95% confidence interval.                               | Intervals based on approximate standard errors.                                      |

| Output                  | Meaning                                                          | Interpretation                                                                                                                                         |
|-------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| sigma_u                 | Estimated standard deviation of the one-sided inefficiency term. | Larger values imply more dispersion in inefficiency.                                                                                                   |
| sigma_v                 | Estimated standard deviation of symmetric random noise.          | Larger values imply more random variation around the frontier.                                                                                         |
| lambda                  | $\sigma_u / \sigma_v$ .                                          | Higher values imply inefficiency variation is large relative to noise variation.                                                                       |
| gamma                   | $\sigma_u^2 / (\sigma_u^2 + \sigma_v^2)$ .                       | Share of total composite-error variance attributed to inefficiency. Values close to 0 imply mostly noise; values close to 1 imply mostly inefficiency. |
| Log Likelihood          | Maximized log-likelihood.                                        | Used to assess fit and compare nested likelihood-based models.                                                                                         |
| AIC and BIC             | Information criteria penalizing model complexity.                | Lower values are preferred when comparing models estimated on the same dataset and dependent variable.                                                 |
| Convergence and message | Optimizer status.                                                | Non-convergence means SFA scores and coefficients should not be treated as stable.                                                                     |

#### 4.4 Cost frontier conventions

For production frontiers, efficiency is naturally reported as a bounded technical-efficiency score. For cost frontiers, two conventions are common. The app supports both so users can match native R/Stata workflows.

| Convention                                        | Best value | Less efficient values | Use when                                                                          |
|---------------------------------------------------|------------|-----------------------|-----------------------------------------------------------------------------------|
| 0-1 reciprocal $E[\exp(-u)   \epsilon]$           | 1          | Closer to 0           | You want cost efficiency on the same 0-1 interpretation as production efficiency. |
| Stata/Frontier cost ratio $E[\exp(u)   \epsilon]$ | 1          | Greater than 1        | You are comparing to a workflow that reports excess cost as a ratio.              |

### 5. Comparing DEA and SFA results

DEA and SFA can produce different rankings because they answer related but distinct questions. DEA is nonparametric and deterministic: all distance from the frontier is treated as inefficiency or slack relative to the selected peer set. SFA is parametric and stochastic: it separates random noise from one-sided inefficiency under distributional assumptions.

| Reason results differ           | Implication for users                                                                                           |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Different frontier construction | DEA uses observed best-practice combinations; SFA estimates a functional form such as Cobb-Douglas or Translog. |
| Noise treatment                 | DEA does not separate random noise from inefficiency. SFA                                                       |



| Reason results differ         | Implication for users                                                                                                                   |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
|                               | does, but only under its distributional assumptions.                                                                                    |
| Multiple outputs              | DEA can use multiple outputs directly. The current SFA module is single-output.                                                         |
| Scale assumptions             | DEA can switch between CRS and VRS. SFA returns-to-scale interpretation depends on estimated coefficients and model form.               |
| Outliers                      | DEA frontiers can be sensitive to extreme observations. SFA can also be affected, but the likelihood distributes variation differently. |
| Variables and transformations | DEA typically uses raw levels. SFA commonly uses logs. Mismatched transformations create different results.                             |

### Recommended comparison practice

Do not interpret disagreement between DEA and SFA as an error by itself. First confirm that variables, transformations, orientation, returns-to-scale assumptions, time handling, and output definitions match the intended comparison.

## 6. Time Trends outputs

The Time Trends tab appears when the selected time column has at least two periods. It summarizes how already-estimated scores change for the same base unit over time. It is a descriptive change analysis, not a Malmquist productivity index and not a panel SFA estimator.

### 6.1 Period summary outputs

| Column                          | Definition                                                       | How to interpret                                                                                                    |
|---------------------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Time_Period                     | Ordered period label.                                            | Numeric and date-like periods are sorted chronologically where possible.                                            |
| Score_Label                     | Readable name of the score summarized.                           | Examples: DEA efficiency, SFA efficiency, Scale efficiency.                                                         |
| Units_With_Score                | Number of units with finite values for that score in the period. | Check for changes in sample composition.                                                                            |
| Mean_Score and Median_Score     | Average and median score in the period.                          | Used to summarize system-level movement.                                                                            |
| Std_Score, Min_Score, Max_Score | Dispersion and range of scores.                                  | Large dispersion may indicate heterogeneity or outliers.                                                            |
| Share_At_Best_Bound             | Share of units at the best bound for the score.                  | For higher-is-better 0-1 scores, this is the share near 1. For cost-ratio SFA, this is the share near 1 from above. |

## 6.2 Unit change outputs

| Column                        | Definition                                                                      | How to interpret                                                                |
|-------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Trend_Unit_ID                 | Base unit identifier used to match observations across time.                    | This is usually the selected DMU identifier, not the combined DMU-period label. |
| First_Period and Last_Period  | The first and last observed periods for the unit and selected score.            | If intermediate periods exist, first-to-last change uses the endpoints.         |
| First_Score and Last_Score    | Score values at the first and last observed periods.                            | Used to calculate raw and directional change.                                   |
| Raw_Change                    | Last_Score minus First_Score.                                                   | Positive means the numeric score rose.                                          |
| Improvement                   | Directional improvement after accounting for whether higher or lower is better. | Positive means improved; negative means declined.                               |
| Improvement_Percent_of_First  | Improvement divided by the absolute first-period score.                         | Useful for comparing proportional changes across units.                         |
| Improved, Declined, Unchanged | Boolean flags based on the improvement value.                                   | Used to calculate shares improved or declined.                                  |

## 7. Malmquist productivity outputs

The Malmquist Productivity Index is a DEA-based measure of productivity change between adjacent periods. It decomposes change into efficiency change, often called catch-up, and technical change, often called frontier shift.

### 7.1 Four DEA distance-style scores

For each unit present in two adjacent periods, the app evaluates the period-0 and period-1 observations against both period frontiers. The four scores are exported because they are useful for auditing and comparison with native DEA libraries.

| Column                          | Meaning                                                       |
|---------------------------------|---------------------------------------------------------------|
| e00_Period0_On_Period0_Frontier | Period-0 observation evaluated against the period-0 frontier. |
| e01_Period0_On_Period1_Frontier | Period-0 observation evaluated against the period-1 frontier. |
| e10_Period1_On_Period0_Frontier | Period-1 observation evaluated against the period-0 frontier. |
| e11_Period1_On_Period1_Frontier | Period-1 observation evaluated against the period-1 frontier. |

```
Efficiency change, EC = e11 / e00
Technical change, TC = sqrt((e10 / e11) * (e00 / e01))
Malmquist MPI, MPI = EC * TC
                = sqrt((e10 / e00) * (e11 / e01))
```

| Output            | Interpretation                                   |
|-------------------|--------------------------------------------------|
| Malmquist_MPI > 1 | Productivity improved from period 0 to period 1. |

| Output                   | Interpretation                               |
|--------------------------|----------------------------------------------|
| Malmquist_MPI = 1        | No measured productivity change.             |
| Malmquist_MPI < 1        | Productivity declined.                       |
| Efficiency_Change_EC > 1 | The unit caught up toward the frontier.      |
| Technical_Change_TC > 1  | The frontier shifted favorably for the unit. |

## 7.2 Malmquist table outputs

| Column                      | Definition                                                           | How to interpret                                                           |
|-----------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------------|
| Period_0 and Period_1       | Adjacent time periods being compared.                                | The app compares each period with the next ordered period.                 |
| Trend_Unit_ID               | Unit present in both adjacent periods.                               | Malmquist is calculated only for common units in the adjacent period pair. |
| Malmquist_MPI               | Overall productivity index.                                          | Above 1 means productivity improvement.                                    |
| Efficiency_Change_EC        | Catch-up component.                                                  | Above 1 means the unit moved closer to the frontier.                       |
| Technical_Change_TC         | Frontier-shift component.                                            | Above 1 means the technology frontier moved favorably.                     |
| Productivity_Change_Percent | $(MPI - 1) * 100$ .                                                  | Percent productivity change between adjacent periods.                      |
| All_Distance_LPs_Succeeded  | Whether all four DEA distance calculations succeeded.                | If FALSE, inspect messages and missing scores.                             |
| LP_Success_Rate             | Success rate of distance LPs in a period-pair summary.               | Low rates often indicate VRS cross-period infeasibility or data problems.  |
| Same_Reference_Units        | Whether frontiers were restricted to units observed in both periods. | Use this for a stricter common-reference comparison.                       |
| MPI_Chain_Index             | Cumulative product of adjacent MPI values starting from 1.           | Shows cumulative productivity movement over more than two periods.         |

### VRS infeasibility caveat

With VRS and cross-period frontiers, some observations may fall outside the feasible technology of another period. The app reports missing MPI values and solver diagnostics rather than forcing an index. CRS Malmquist or common-reference settings may be preferable for some publication workflows.

## 8. Bootstrap, robustness, and sensitivity outputs

The app includes robustness diagnostics that help users understand how sensitive scores and rankings are to sampling variation, frontier resampling, or influential observations. These outputs are useful screening tools, but they are not a substitute for a dedicated inference design tailored to the study question.

## 8.1 DEA bootstrap outputs

| Column                                  | Definition                                                       | How to interpret                                                         |
|-----------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------|
| DEA_Bootstrap_Mean                      | Mean DEA score across bootstrap frontier-resampling draws.       | Compare with baseline DEA_Efficiency to assess sensitivity.              |
| DEA_Bootstrap_Std_Error                 | Standard deviation of bootstrap scores.                          | Higher values imply less stable score estimates.                         |
| DEA_Bootstrap_Bias                      | Bootstrap mean minus baseline score.                             | Positive or negative bias indicates systematic bootstrap displacement.   |
| DEA_Bootstrap_Bias_Corrected            | Simple bias-corrected score.                                     | Use as a sensitivity indicator, not a definitive corrected DEA estimate. |
| DEA_Bootstrap_CI_Lower / Upper          | Percentile interval bounds at the selected confidence level.     | Wide intervals indicate uncertainty or frontier sensitivity.             |
| DEA_Bootstrap_Frontier_Rate             | Share of bootstrap draws in which the unit is near the frontier. | High values suggest robust frontier status.                              |
| DEA_Bootstrap_Rank_Mean / Lower / Upper | Bootstrap rank summary.                                          | Useful when rankings are more important than absolute scores.            |
| DEA_Bootstrap_LP_Success_Rate           | Share of bootstrap LPs solved successfully.                      | Low values indicate feasibility or numerical problems.                   |

## 8.2 SFA bootstrap outputs

| Output table             | Contents                                                                                                    | Use                                                                   |
|--------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| SFA efficiency bootstrap | Baseline efficiency, bootstrap mean, standard deviation, percentile interval, bias, and valid replications. | Assess uncertainty in fitted SFA efficiency for each DMU.             |
| SFA parameter bootstrap  | Coefficient bootstrap summaries and percentile intervals.                                                   | Assess whether coefficient estimates are stable under row resampling. |
| SFA metric bootstrap     | Bootstrap summaries for sigma_u, sigma_v, lambda, gamma, log-likelihood, AIC, and BIC.                      | Assess stability of variance decomposition and model fit.             |

## 8.3 Leave-one-out robustness

| Column                                       | Definition                                                                          | How to interpret                                                         |
|----------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Omitted_DMU_ID                               | DMU removed from the reference set for the jackknife run.                           | Each row evaluates the influence of one omitted DMU.                     |
| Mean_Absolute_Change_in_Remaining_DEA_Scores | Average absolute DEA score change among all remaining DMUs after omitting the unit. | Large values identify influential benchmark units or potential outliers. |
| Baseline_DEA_Efficiency                      | Original DEA score of the omitted unit.                                             | Influential units are often, but not always, frontier units.             |

## 9. Diagnostics, limitations, and recommended reporting

### 9.1 Diagnostics to check before interpreting results

- Confirm the selected inputs and outputs are conceptually aligned with the production process.
- Confirm model variables are in consistent units and do not mix rates with counts unless intended.
- Inspect rows removed due to missing, non-numeric, or non-positive model values.
- Check whether the number of DMUs is adequate for the number of selected inputs and outputs.
- Review DEA solver statuses and messages for missing scores or infeasible models.
- Review SFA convergence before interpreting coefficients or efficiency scores.
- Check whether repeated DMU-time data are being pooled, analyzed by period, summarized as time trends, or evaluated with Malmquist.

### 9.2 Important limitations

| Area                        | Limitation                                                          | Recommended response                                                                                              |
|-----------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| DEA                         | Sensitive to peer-set definition, outliers, and omitted variables.  | Use group filters, inspect influential units, and run robustness checks.                                          |
| DEA targets                 | Targets are relative to observed peers and selected variables only. | Treat as benchmark scenarios, not automatic prescriptions.                                                        |
| SFA                         | Current estimator is pooled cross-sectional half-normal SFA.        | Do not interpret it as xtfreight, true fixed-effects SFA, truncated-normal SFA, or an inefficiency-effects model. |
| SFA standard errors         | Optimizer-Hessian standard errors are approximate.                  | Use bootstrap output or a dedicated statistical package for publication-grade inference.                          |
| Time Trends                 | Describes score changes; does not decompose productivity.           | Use Malmquist for DEA productivity decomposition.                                                                 |
| Malmquist                   | VRS cross-period distance functions can be infeasible.              | Check LP success rates and consider CRS or common-reference settings.                                             |
| Comparisons across software | R/Stata defaults and score conventions may differ.                  | Match orientation, returns, variables, logging, cost convention, and time handling.                               |

### 9.3 Suggested reporting language

The following template can be adapted for reports, methods appendices, or stakeholder briefings.

Efficiency was assessed using Data Envelopment Analysis (DEA) and Stochastic Frontier Analysis (SFA). DEA scores were estimated using the selected input/output variables, orientation, and returns-to-scale assumption. DEA\_Efficiency is reported on a 0-1 scale, where 1 indicates that a unit lies on the estimated DEA frontier. For output-oriented DEA, the underlying output expansion factor is exported separately as DEA\_Radial\_Factor.

SFA scores were estimated using a pooled half-normal stochastic frontier. SFA\_Efficiency is interpreted as the expected efficiency conditional on the model residual and the selected cost/production convention. DEA and SFA results may differ because DEA is deterministic and nonparametric, while SFA is parametric and separates random noise from one-sided inefficiency.

Where repeated time periods were available, the Time Trends tab summarized first-to-last score changes for the same unit. Malmquist productivity indices were estimated separately using adjacent-period DEA distance functions; MPI values above 1 indicate productivity improvement, with efficiency-change and technical-change components reported separately.

## 10. Output column dictionary

This section summarizes the most common exported columns. Exact columns vary depending on selected options, model settings, and whether bootstrap, cross-efficiency, time trends, or Malmquist modules are enabled.

| Family            | Columns                                                                                                              | Plain-language meaning                                               |
|-------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Identification    | DMU_ID, Original_Row, selected DMU/time/group/environmental columns                                                  | Unit identity and preserved metadata.                                |
| DEA primary       | DEA_Efficiency, DEA_Efficiency_0_1, DEA_Radial_Factor, DEA_Rank, DEA_Status, DEA_Message                             | Main DEA score, radial factor, rank, and LP diagnostics.             |
| DEA compatibility | DEA_R_Benchmarking_Style_Score, DEA_R_Benchmarking_Style_Rank                                                        | Columns intended to ease comparison with native R/Stata conventions. |
| DEA targets       | Current_Input_*, Target_Input_*, Total_Input_Reduction_*, Current_Output_*, Target_Output_*, Total_Output_Increase_* | Current values, peer-implied targets, and improvement quantities.    |
| DEA peers         | DEA_Peers, Peer_DMU_ID, Lambda_Weight                                                                                | Reference units and peer weights.                                    |
| Scale             | DEA_CRS_Efficiency, DEA_VRS_Efficiency, Scale_Efficiency, CRS_Lambda_Sum, Returns_to_Scale                           | Scale diagnostics and returns-to-scale classification.               |
| Benchmark support | Reference_Count, Reference_Weight_Sum, Benchmark_Support_Score                                                       | How often a DMU is used as a benchmark by other units.               |
| Cross-efficiency  | CCR_Cross_Efficiency_Mean, Min, Max, CCR_Self_Efficiency_from_Multiplier, Self_vs_Cross_Gap                          | Multiplier-weight peer appraisal under CCR/CRS.                      |
| Super-efficiency  | DEA_SuperEfficiency, DEA_SuperEfficiency_Rank, DEA_SuperEfficiency_Status                                            | Leave-one-out frontier ranking for efficient units.                  |
| SFA primary       | SFA_Efficiency, SFA_Rank, SFA_Model_Input_Columns, SFA_Model_Output_Column, SFA_Variables_Already_Logged             | Main SFA score, rank, and model variable audit fields.               |

| Family               | Columns                                                                                     | Plain-language meaning                                                    |
|----------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| SFA coefficients     | Parameter, Coefficient, Std_Error, t_stat, p_value, CI_Lower_95%, CI_Upper_95%, Elasticity  | Frontier coefficient table and approximate inference.                     |
| SFA variance and fit | sigma_u, sigma_v, lambda, gamma, log_likelihood, AIC, BIC, convergence message              | Noise/inefficiency decomposition and fit diagnostics.                     |
| Time Trends          | Period summary, unit change, score long, change summary outputs                             | Descriptive first-to-last score changes and period-level score summaries. |
| Malmquist            | Malmquist_MPI, Efficiency_Change_EC, Technical_Change_TC, e00, e01, e10, e11, chain indices | DEA productivity change and decomposition across adjacent periods.        |
| Bootstrap            | DEA_Bootstrap_*, SFA_Boot_*                                                                 | Resampling-based robustness and uncertainty summaries.                    |

## 11. Glossary

| Term                   | Definition                                                                                                                  |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Best-practice frontier | The estimated boundary representing the highest observed or modeled performance for the selected input-output relationship. |
| CRS                    | Constant returns to scale: scaling all inputs proportionally is assumed to scale outputs proportionally.                    |
| DEA                    | Data Envelopment Analysis, a nonparametric linear-programming frontier method.                                              |
| DMU                    | Decision making unit: the entity being evaluated.                                                                           |
| DRS                    | Decreasing returns to scale: larger scale is associated with less than proportional output expansion.                       |
| EC                     | Efficiency change in Malmquist analysis; the catch-up component.                                                            |
| IRS                    | Increasing returns to scale: larger scale is associated with more than proportional output expansion.                       |
| Lambda weights         | DEA peer weights used to construct a target point on the reference frontier.                                                |
| MPI                    | Malmquist Productivity Index; a DEA-based adjacent-period productivity change measure.                                      |
| RTS                    | Returns to scale.                                                                                                           |
| SFA                    | Stochastic Frontier Analysis, a parametric frontier method that separates random noise from inefficiency.                   |
| Slack                  | Non-radial input excess or output shortfall after the radial efficiency movement.                                           |

| Term | Definition                                                                                                  |
|------|-------------------------------------------------------------------------------------------------------------|
| TC   | Technical change in Malmquist analysis; the frontier-shift component.                                       |
| VRS  | Variable returns to scale: the frontier allows scale-adjusted comparisons through the convexity constraint. |

## 12. Classic references for users who want more detail

- Aigner, D., Lovell, C. A. K., and Schmidt, P. (1977). Formulation and estimation of stochastic frontier production function models. *Journal of Econometrics*.
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- Charnes, A., Cooper, W. W., and Rhodes, E. (1978). Measuring the efficiency of decision making units. *European Journal of Operational Research*.
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- Simar, L., and Wilson, P. W. (1998). Sensitivity analysis of efficiency scores: How to bootstrap in nonparametric frontier models. *Management Science*.
- Simar, L., and Wilson, P. W. (2007). Estimation and inference in two-stage, semi-parametric models of production processes. *Journal of Econometrics*.



## Quick interpretation checklist

| Check       | Question to answer before reporting                                                                                                              |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Model setup | Are the selected DMUs comparable? Are inputs and outputs conceptually correct? Are transformations aligned with the intended DEA/SFA comparison? |
| DEA score   | Is the orientation input or output? Are you interpreting the bounded score or the radial factor?                                                 |
| Targets     | Do the proposed target reductions/increases make operational sense after reviewing peers and slacks?                                             |
| Scale       | Is low efficiency due to pure technical inefficiency, scale inefficiency, or both?                                                               |
| SFA         | Did the model converge? Is gamma plausible? Are variables raw/logged as intended?                                                                |
| Time        | Are results pooled over time, separate by time period, descriptive trends, or Malmquist productivity measures?                                   |
| Uncertainty | Do bootstrap intervals, jackknife influence, or LP success diagnostics suggest instability?                                                      |
| Reporting   | Are assumptions, orientation, returns to scale, variables, time handling, and software version documented?                                       |